

Recitation 8 Spring 2023 (Student Version)

Heaps, B-Trees, Red-Black Trees

1. [5 minutes] True/False

1. In a 2-3-4 Tree, insertion can be performed on any node so long as the other nodes are adjusted accordingly
2. Deleting the highest priority element and reorganizing the heap will only take $O(\log n)$ time.
3. A red-black tree may have all red nodes.
4. The following array is a max-heap:

Int maxHeap[] = {10,5,3,4,1};

2. [5 minutes] Short Answer

1. What are the steps for Depth-First Traversal in a Binary Tree?

2. Consider an R-B tree with a black height of 5. In any given path from root to leaf, give both the minimum and maximum amount of total nodes within the path (Root = 1 Node, Leaf = 1 Node).

3. [5 minutes] What is the max height of a Red-Black Tree with 6 nodes? Draw an example of this Red-Black Tree.

4. [7 minutes] Insert 2, 1, 4, 5, 9, 3, 6, 7 into an initially empty red-black tree

[Red/Black Tree Visualization \(usfca.edu\)](https://www.usfca.edu/~cs/faculty/robertson/RedBlackTreeVisualization.html)

5. [15 minutes] Write a recursive function to check whether or not a given complete tree represents a max heap. You are permitted to use the following methods:

getLeftChild(), getRightChild(), getData(). Use the following function declaration and assume the first call will pass in the root for the Node argument and -1 for the parentData argument.

```
public boolean isHeap(Node root, int parentData){
```

6. **[10 Minutes]** Fill in the chart below with the worst-case time complexity for each of the following operations:

	Search	Insert	Delete
B-Trees			
AVL Tree			
Red Black Tree			
	Peek	Push	Pop
Heap			